AKASH MAHAPATRA

AI/ML Developer | Full-Stack Developer | Computer Vision

Machine Learning · Geospatial AI · Intelligent Systems · Automation

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

B.Tech Artificial Intelligence & Machine Learning student with hands-on experience developing AI, computer-vision, remote-sensing and full-stack applications. Built projects involving satellite-image analysis, hybrid AI pipelines, Java-based enterprise applications, secure REST APIs and relational databases. Skilled in integrating machine learning, backend services and user interfaces to develop practical solutions for real-world problems.

EDUCATION

B.Tech in Artificial Intelligence & Machine Learning *2024 – 2028*

Techno Bengal Institute of Technology, Kolkata

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, SQL

AI/ML & Deep Learning: Machine Learning, Deep Learning, Scikit-learn, PyTorch, CNNs, Transformers, Sentence Embeddings

Computer Vision & Geospatial AI: Computer Vision, SegFormer, Grounding DINO, GeoTIFF, Optical/SAR Analysis, Spectral Indices

Frontend Development: React.js, Next.js, Java Swing

Backend Development & Security: FastAPI, Spring Boot, Spring Security, REST APIs, JWT

Databases & ORM: MySQL, PostgreSQL, Supabase, Spring Data JPA, Hibernate

DevOps, Automation & Tools: Docker, Render, n8n, Webhooks, Git, GitHub, Maven, VS Code

PROJECTS

Employee Payroll Management System – Java Full-Stack Application | Live Demo | GitHub

Java 17 · Swing · Spring Boot · Spring Security · REST APIs · MySQL · JPA/Hibernate

- Developed a Java desktop payroll application with a Swing GUI, Spring Boot REST backend and MySQL database for employee management, salary calculations and payroll summaries.
- Implemented secure registration and login using Spring Security, BCrypt password hashing and JWT authentication, with protected REST endpoints and persistent employee records.
- Applied OOP principles to full-time and part-time salary calculations; implemented employee CRUD, search, validation and payroll analytics, with 11 passing automated tests recorded.

SatQuery AI – Remote Sensing & Geospatial AI Platform | Live Demo | GitHub

Python · Computer Vision · Deep Learning · Remote Sensing · GeoTIFF · Optical/SAR

- Developed a multimodal satellite-image analysis workflow supporting image understanding, land-cover analysis, object localization and before–after comparison.
- Worked with SegFormer, Grounding DINO, MLP/CNN-based water analysis and Siamese CNN workflows with spectral-index and evidence-based verification methods.
- Designed task-dependent routing across image-analysis methods based on the input and remote-sensing query.

Stratify.AI – Hybrid AI Startup Intelligence Platform | Live Demo | GitHub

Python · FastAPI · Scikit-learn · MiniLM · Supabase · PostgreSQL · LLM APIs · Render

- Built a hybrid AI platform combining deterministic financial analytics, rule-based scoring, machine learning and LLM reasoning for startup assessment.
- Developed an ML intent-classification workflow and a MiniLM embedding-based mentor recommendation system using semantic similarity.
- Built an analysis-aware Founder Copilot using startup-report context, intent routing and structured AI responses; deployed frontend and backend services on Render.

ACHIEVEMENTS & HACKATHONS

- **3rd Position – Internal Smart India Hackathon (SIH):** Secured third position in the college-level internal SIH competition.
- **4th Position – GenFusion:** Secured fourth position with **Stratify.AI**, a hybrid AI startup intelligence platform.
- **5th Position – Electronovation:** Secured fifth position in the technical innovation competition.
- **Hackathon Participation:** Actively participated in inter-college and external hackathons, gaining practical experience in collaborative problem-solving and rapid prototyping.

ADDITIONAL INFORMATION

Languages: English (Professional) | Bengali (Native) | Hindi (Native)